Chapter 9
Sets

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What we will learn

- Translate membership, inclusion, and equality into logical statements.
- Prove statements about unions, intersections, and complements.
- Keep the ambient type explicit, including for empty and universal sets.
- Treat sets as inputs and outputs of functions.
- Prove that no function $X \to \mathcal{P}(X)$ can be surjective.

The organizing idea

A set in a type X is specified by a predicate on X. To ask whether x belongs to the set is to ask whether that predicate holds at x.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 9; Sections 9.1–9.3.

9.1: Membership evaluates a predicate

For a predicate P on X, write

$$S = \{x : X \mid P(x)\}, \qquad a \in S \iff P(a).$$

The notation $a \notin S$ means $\neg(a \in S)$.

A concrete set of integers

Let $S = \{n : \mathbb{Z} \mid n \leq 3\}$. Then $1 \in S$ because $1 \leq 3$, while $7 \notin S$ because $7 \not\leq 3$. Negative integers can also belong to S: for example, $-10 \leq 3$.

A set defined by parity

For $O = \{n : \mathbb{N} \mid n \text{ is odd}\}$, we have $10 \notin O$: $10 = 2 \cdot 5$ is even, and no natural number is both even and odd.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.1.1–9.1.2.

Inclusion is a universal implication

Definition

For sets U, V in the same type X,

$$U \subseteq V \quad \iff \quad \forall x \in X, \quad x \in U \implies x \in V.$$

Claim. $\{a : \mathbb{N} \mid 4 \mid a\} \subseteq \{b : \mathbb{N} \mid 2 \mid b\}$.

Proof. Choose an arbitrary $a \in \mathbb{N}$ with $4 \mid a$. There is a $k \in \mathbb{N}$ with $a = 4k$. Now

$$a = 4k = 2(2k).$$

The witness $2k \in \mathbb{N}$ proves $2 \mid a$. Therefore every element of the first set belongs to the second.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.1.3.

To refute inclusion, find an element that escapes

$$U \not\subseteq V \quad \Longleftrightarrow \quad \exists x \in X, \quad x \in U \ \wedge \ x \notin V.$$

For $U = \{x : \mathbb{R} \mid 0 \leq x^2\}$ and $V = \{x : \mathbb{R} \mid 0 \leq x\}$, the witness $x = -3$ works: $0 \leq (-3)^2 = 9$, but $-3 < 0$.

An element in just one of two sets also proves the sets unequal. For example,

$$\{n : \mathbb{N} \mid 4 \mid n\} \neq \{n : \mathbb{N} \mid 2 \mid n\}.$$

Indeed, $6 = 2 \cdot 3$, so 6 belongs to the second set. If $6 = 4k$ with $k \in \mathbb{N}$, then either $k \leq 1$ gives $4k \leq 4$, or $k \geq 2$ gives $4k \geq 8$. Thus 6 is absent from the first set.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.1.4 and 9.1.6.

Set equality means the same membership test

Extensionality

$U = V$ if and only if $\forall x \in X,\ x \in U \iff x \in V$. Equivalently, prove both $U \subseteq V$ and $V \subseteq U$.

Example. The sets

$$O = \{x : \mathbb{Z} \mid x \text{ is odd}\}, \qquad M = \{x : \mathbb{Z} \mid \exists k \in \mathbb{Z},\ x = 2k - 1\}$$

are equal. Fix $x \in \mathbb{Z}$.

- If $x \in O$, write $x = 2l + 1 = 2(l + 1) - 1$. The integer witness $k = l + 1$ gives $x \in M$.
- If $x \in M$, write $x = 2k - 1 = 2(k - 1) + 1$. The integer witness $l = k - 1$ shows that x is odd.

Both implications hold for every x, hence $O = M$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.1.5.

Finite sets turn membership into alternatives

For a fixed ambient type,

$$x \in \{a, b\} \quad \Longleftrightarrow \quad x = a \ \vee \ x = b.$$

Order and repetitions do not affect a set: $\{a, b, a\} = \{b, a\}$.

A complete solution set

We claim $\{x : \mathbb{R} \mid x^2 - x - 2 = 0\} = \{-1, 2\}$. For an arbitrary real x:

- If $x^2 - x - 2 = 0$, then $(x + 1)(x - 2) = 0$. Since $\mathbb{R}$ has no zero divisors, $x + 1 = 0$ or $x - 2 = 0$, giving $x = -1$ or $x = 2$.
- Conversely, substituting $x = -1$ gives $1 + 1 - 2 = 0$; substituting $x = 2$ gives $4 - 2 - 2 = 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.1.8.

Lean: Strip off the set notation

```
example : Set.Subset
    {a : Nat | Exists (fun k : Nat => a = 4 * k)}
    {b : Nat | Exists (fun j : Nat => b = 2 * j)} := by
  intro a ha
  cases ha with
  | intro k hk =>
      use 2 * k
      calc a = 4 * k := hk
        _ = 2 * (2 * k) := by ring
```

Here `Exists` spells out the divisibility witness.

Membership	`dsimp` exposes the predicate.
Subset	`intro x hx` starts the implication.
Set equality	`ext x` creates a membership equivalence.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.1.1, 9.1.3, and 9.1.5; textbook-environment demonstration.

Practice 9.1: Do the sets contain exactly what is claimed?

1. Prove $\{n : \mathbb{N} \mid 20 \mid n\} \subseteq \{n : \mathbb{N} \mid 5 \mid n\}$. Is the reverse inclusion true?
2. Prove or disprove
$$\{t : \mathbb{R} \mid t^2 - 5t + 4 = 0\} = \{4\}.$$
3. Decide whether -3 belongs to $\{x : \mathbb{R} \mid \forall y \in \mathbb{R},\ x \leq y^2\}$.

State the logical form of each task before proving it. An equation satisfied by every listed element does not prove the list is complete.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 9.1.10.

Practice 9.1: Solution ideas

1. If $n = 20k$, then $n = 5(4k)$, so $5 \mid n$. The reverse inclusion fails at $n = 5$: it is a multiple of 5 but not of 20.
2. The equality is false. The witness $t = 1$ satisfies $1 - 5 + 4 = 0$, but $1 \notin \{4\}$. In fact, $(t-1)(t-4) = t^2 - 5t + 4$, so the full solution set is $\{1, 4\}$.
3. Yes. For every real y, $y^2 \geq 0$, hence $-3 \leq 0 \leq y^2$. The same input -3 works for *all* values quantified inside the predicate.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 9.1.10.

9.2: Set operations package familiar logical operations

Fix sets U, V in a type X and an element $x \in X$.

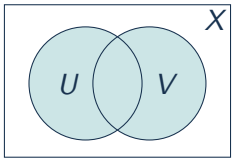
Set	Membership condition	Logical operation
$U \cup V$	$x \in U \vee x \in V$	or
$U \cap V$	$x \in U \wedge x \in V$	and
U^c	$x \notin U$	not
$\emptyset$	false	false
univ_X	true	true

The complement is relative to X: the complement of $\{0\}$ in $\mathbb{N}$ contains no negative numbers; its complement in $\mathbb{Z}$ does.

We often write X for its universal set on paper. Lean keeps the type `X` and the set `Set.univ` separate.

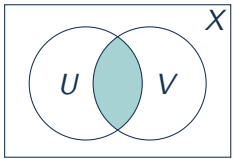
Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.2.1, 9.2.3, and 9.2.5–9.2.7.

Union accepts either membership; intersection requires both



$U \cup V$

To prove membership, establish at least one side. From membership, be ready for two cases.



$U \cap V$

To prove membership, establish both sides. From membership, obtain two usable facts.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.2.1 and 9.2.3; original diagrams.

A worked intersection proof uses both hypotheses

Claim.

$$\{-2, 3\} \cap \{x : \mathbb{Q} \mid x^2 = 9\} \subseteq \{x : \mathbb{Q} \mid 0 < x\}.$$

Proof. Fix $t \in \mathbb{Q}$ in the intersection. Then

$$(t = -2 \ \vee \ t = 3) \qquad \text{and} \qquad t^2 = 9.$$

- If $t = -2$, then $t^2 = 4$. Together with $t^2 = 9$, this gives $4 = 9$, a contradiction.
- If $t = 3$, then $0 < t$.

The first condition narrows the possibilities; the second rules out the unwanted one. Thus every element of the intersection is positive.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.2.3.

Empty sets and universal sets require different proofs

An empty intersection. Let $A = \{n : \mathbb{Z} \mid n \equiv 1 \pmod 5\}$ and $B = \{n : \mathbb{Z} \mid n \equiv 2 \pmod 5\}$. If $x \in A \cap B$, then $x = 1 + 5k = 2 + 5l$ for some integers k, l. Thus $1 = 5(k - l)$, impossible. Hence $A \cap B = \emptyset$.

A universal union. Let $C = \{x : \mathbb{R} \mid -1 < x\}$ and $D = \{x : \mathbb{R} \mid x < 1\}$. For an arbitrary $t \in \mathbb{R}$:

- if $t \leq 0$, then $t < 1$, so $t \in D$;
- if $t > 0$, then $-1 < t$, so $t \in C$.

Therefore every real number belongs to $C \cup D$, and $C \cup D = \mathbb{R}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.2.6–9.2.7.

Lean: Arithmetic facts versus propositional structure

```
example : Set.union ({1, 2} : Set Nat) {2, 4}
    = {1, 2, 4} := by
  ext n
  dsimp
  exhaust
```

After unfolding, the goal is the logical equivalence

$$((n = 1 \lor n = 2) \lor (n = 2 \lor n = 4)) \iff (n = 1 \lor n = 2 \lor n = 4).$$

The book's `exhaust` handles this propositional rearrangement. It does not replace a needed arithmetic argument such as proving two residue conditions incompatible.

For an intersection hypothesis, split the conjunction; for a union hypothesis, split into cases. These are the same proof patterns used before sets were introduced.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.2.2; Exercises 9.2.8.

Practice 9.2: Calculate, then prove a general identity

1. In the ambient type $\mathbb{N}$, simplify

$$\{0,1,2,3,4\} \cap \{0,2,4,6,8\}, \qquad \{3,4,5\}^c \cap \{1,3,5,7,9\}.$$

2. For arbitrary sets U, V in the same type X, prove

$$(U \cup V)^c = U^c \cap V^c.$$

Start with an arbitrary $x \in X$ and translate each side into logic.

The picture can suggest an answer. A membership proof establishes it for every element.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 9.2.8; original general-identity exercise.

Practice 9.2: Solution ideas

The explicit intersections are $\{0, 2, 4\}$ and $\{1, 7, 9\}$. In the second expression, remove 3 and 5 from the listed set; 4 was absent already.

For the identity, fix $x \in X$.

- Suppose $x \notin U \cup V$. If $x \in U$, it would belong to the union; similarly for V. Hence $x \notin U$ and $x \notin V$, so $x \in U^c \cap V^c$.
- Suppose $x \in U^c \cap V^c$. Then $x \notin U$ and $x \notin V$. Either case of $x \in U \cup V$ contradicts one of these facts. Thus $x \notin U \cup V$.

The membership conditions are equivalent for every x, so extensionality gives the set equality.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 9.2 definitions; solutions to the preceding practice.

9.3: A set can itself be an element

The type of all sets in X is written $\mathcal{P}(X)$ on paper and `Set X` in Lean.

Object	Type
3	$\mathbb{N}$
$\{3, 4\}$	$\mathcal{P}(\mathbb{N})$
$\{\{3, 4\}, \{4, 5, 6\}\}$	$\mathcal{P}(\mathcal{P}(\mathbb{N}))$
$\mathcal{F} = \{S : \mathcal{P}(\mathbb{N}) \mid 3 \in S\}$	$\mathcal{P}(\mathcal{P}(\mathbb{N}))$

For the set E of even natural numbers, $E \notin \mathcal{F}$ because $3 \notin E$. However, $\{3, 4\} \in \mathcal{F}$ because $3 \in \{3, 4\}$.

Membership $S \in \mathcal{F}$ and inclusion $S \subseteq T$ are different statements about different types.

Source: Heather Macbeth, *The Mechanics of Proof*, Definition 9.3.1; Example 9.3.2.

A function on sets can lose information

Define $p : \mathcal{P}(\mathbb{N}) \to \mathcal{P}(\mathbb{N})$ by

$$p(S) = \{n : \mathbb{N} \mid n + 1 \in S\}.$$

For instance, $p(\{0, 2, 4\}) = \{1, 3\}$.

Claim: p is not injective

Take $S = \{0\}$ and $T = \emptyset$. Since $n + 1 > 0$ for every $n \in \mathbb{N}$, no n satisfies $n + 1 \in S$. Thus $p(S) = \emptyset = p(T)$. But $S \neq T$, because $0 \in S$ and $0 \notin T$.

The output never tests whether $0 \in S$; all tested inputs have the form $n + 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.3.3.

The same rule over integers preserves information

Define $q : \mathcal{P}(\mathbb{Z}) \to \mathcal{P}(\mathbb{Z})$ by $q(S) = \{n : \mathbb{Z} \mid n + 1 \in S\}$.

Claim. q is injective. Suppose $q(S) = q(T)$. To prove $S = T$, fix an arbitrary $k \in \mathbb{Z}$. Then $k - 1$ is also an integer, and

$$\begin{aligned} k \in S &\iff (k-1) + 1 \in S \\ &\iff k - 1 \in q(S) \\ &\iff k - 1 \in q(T) \\ &\iff (k-1) + 1 \in T \\ &\iff k \in T. \end{aligned}$$

Extensionality gives $S = T$. The ability to use $k - 1$ is essential.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.3.4.

Cantor's construction: A set that differs on the diagonal

Theorem

For every type X, no function $f : X \to \mathcal{P}(X)$ is surjective.

Define $D = \{x : X \mid x \notin f(x)\}$. The membership of x in D is the opposite of its membership in $f(x)$.

	a	b	c	d
$f(a)$	**1**	0	1	0
$f(b)$	1	**0**	0	1
$f(c)$	0	1	**1**	1
$f(d)$	1	0	0	**0**
D	0	1	0	1

In this illustrative finite table, 1 means membership and 0 means nonmembership.

D differs from row $f(a)$ at column a, from row $f(b)$ at column b, and so on. The definition works without enumerating X.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.3.5; original membership matrix.

Cantor's proof: The proposed preimage contradicts itself

Assume $f : X \to \mathcal{P}(X)$ is surjective and set $D = \{x : X \mid x \notin f(x)\}$. Surjectivity supplies $d \in X$ with $f(d) = D$. Consequently,

$$d \in D \iff d \notin f(d) \iff d \notin D.$$

- If $d \in D$, the defining condition gives $d \notin f(d) = D$, a contradiction.
- If $d \notin D$, then $d \notin f(d)$ because $f(d) = D$. By the definition of D, this says $d \in D$, again a contradiction.

Thus the assumption of surjectivity is impossible.

The obstruction applies to every proposed function f; the diagonal set is built from that very function.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.3.5.

Lean: The diagonal set is a predicate

```
example {X : Type} (f : X -> Set X) : Not (Function.Surjective f) := by
  intro hf
  let D : Set X := fun x => Not ((f x) x)
  cases hf D with
  | intro d hd =>
      by_cases h : D d
      case pos =>
        have hn : Not ((f d) d) := h
        rw [hd] at hn
        exact hn h
      case neg =>
        have hn : Not (Not ((f d) d)) := h
        rw [hd] at hn
        exact hn h
```

Here `D d` means $d \in D$, since a set is represented by a predicate.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.3.5; textbook-environment demonstration.

Practice 9.3: Sets as function values and sequence terms

1. Define $r : \mathcal{P}(\mathbb{N}) \to \mathcal{P}(\mathbb{N})$ by $r(S) = S \cup \{3\}$. Prove that r is not injective.
2. Define a sequence of sets of integers by

$$U_0 = \mathbb{Z}, \qquad U_{n+1} = \{x : \mathbb{Z} \mid \exists y \in U_n,\ x = 2y\}.$$

   Prove by induction that $U_n = \{x : \mathbb{Z} \mid 2^n \mid x\}$ for every $n \in \mathbb{N}$.

In the second task, the induction hypothesis is an equality of sets. Use it to replace a membership condition, then translate divisibility into a witness equation.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 9.3.6.

Practice 9.3: Solution ideas

The union function. The distinct inputs $\emptyset$ and $\{3\}$ both map to $\{3\}$. They are distinct because only the second contains 3.

The sequence. For $n = 0$, $2^0 = 1$ divides every integer, so the formula gives $U_0 = \mathbb{Z}$. Assume the formula for U_n and fix $x \in \mathbb{Z}$.

- If $x \in U_{n+1}$, write $x = 2y$ with $y \in U_n$. The induction hypothesis gives $y = 2^n k$ for some $k \in \mathbb{Z}$. Then $x = 2^{n+1} k$.
- If $x = 2^{n+1} k$, set $y = 2^n k$. Then $y \in U_n$ by the induction hypothesis and $x = 2y$, so $x \in U_{n+1}$.

These implications prove the required set equality at $n + 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 9.3.6.

Chapter 9 review: Expose the underlying logic

Set statement	Proof obligation
$x \in \{t \mid P(t)\}$	Prove $P(x)$.
$U \subseteq V$	Start with arbitrary $x \in U$; prove $x \in V$.
$U = V$	Prove both directions of membership.
$U \neq V$	Find an element in exactly one set.
$U = \emptyset$	Rule out an arbitrary proposed member.
$U = \mathrm{univ}_X$	Show every $x \in X$ belongs.
$S \in \mathcal{F}$	Apply the family's predicate to the set S.

Exit check. For $\mathcal{F} = \{S : \mathcal{P}(\mathbb{N}) \mid 0 \in S\}$, explain why $\{0\} \in \mathcal{F}$ is well-typed and why $0 \in \mathcal{F}$ is not.

Source: Heather Macbeth, *The Mechanics of Proof*, Sections 9.1–9.3.

Reading and code context

Heather Macbeth, *The Mechanics of Proof*, Chapter 9:

- 9.1 Introduction
- 9.2 Set operations
- 9.3 The type of sets

`https://hrmacbeth.github.io/math2001/09_Sets.html`

These notes use original explanations, diagrams, and practice solutions around the cited material. Lean examples use the textbook's teaching environment. In ASCII listings, `Set.Subset` names inclusion, `Set.union` names union, and `Exists` introduces an existential predicate.